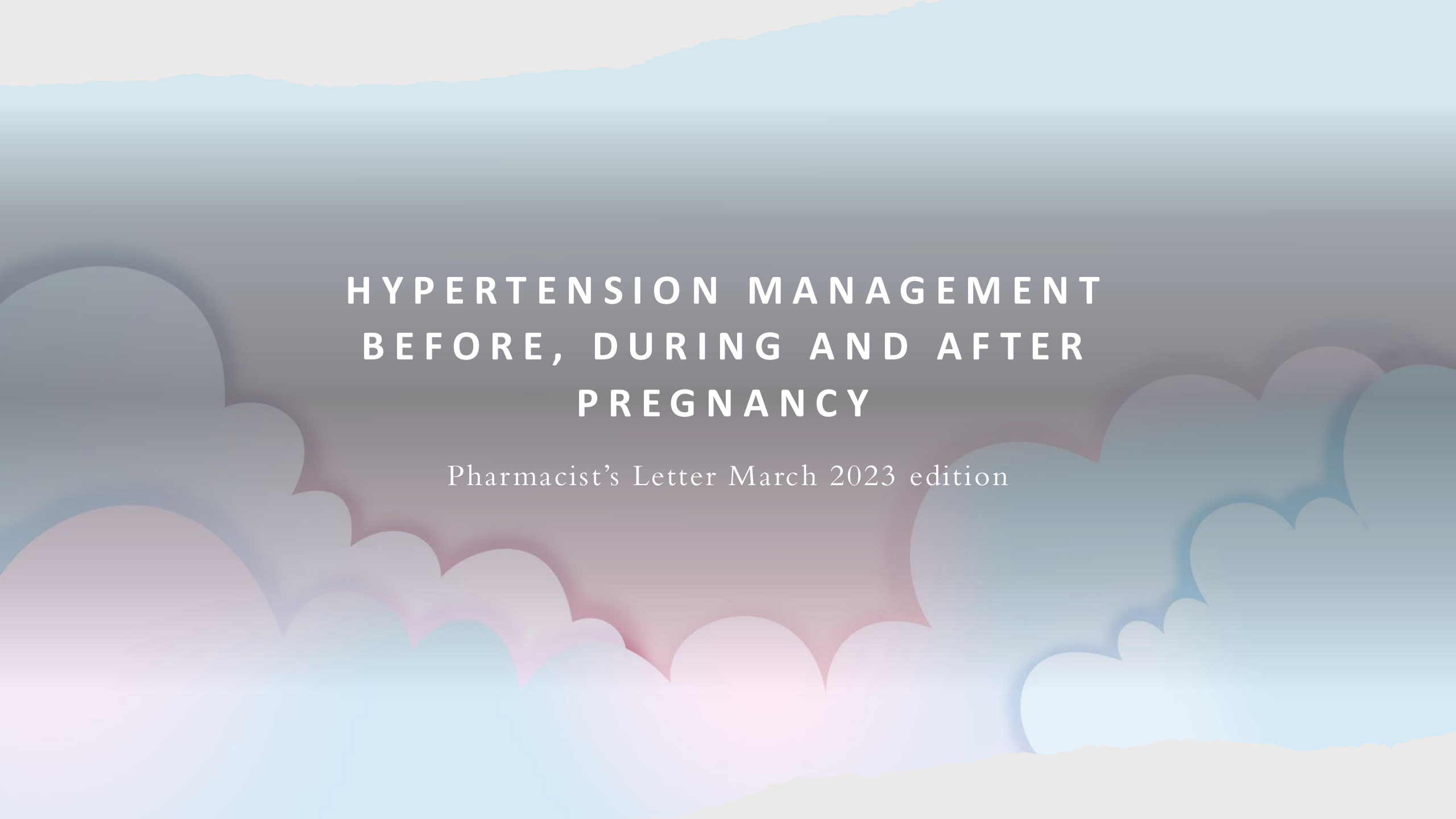




HYPERTENSION MANAGEMENT BEFORE, DURING AND AFTER PREGNANCY

Pharmacist's Letter March 2023 edition

INTRODUCTION:

Before-

About 50% pregnancies are unplanned pregnancies, and patient with hypertension who are on medications with non-safety pregnancy profile loss the pregnancy due to effect of medications on embryo.

During-

Most of the Hypertension medications do not have robust, randomized and controlled trials data to show prevention of Preeclampsia, Preterm birth or infant mortality. Therefore, old medications for Hypertension are being used as safety measure with history of little to no side effects on fetus.

After-

There is very less good quality evidence data available to treat HTN for lactating mothers. In general, highly protein bound, and low lipid soluble medications consider to be safe and assumed not to get transfer to babies via breastmilk.



BACKGROUND: SBP ≥ 140 MM/HG AND DBP ≥ 90 MM/HG

- Why it's important to treat Hypertension during pregnancy?

Lead cause for –

Eclampsia

HELLP Syndrome

Low birth weight

Poor fetal growth

Preterm Birth

NICU care

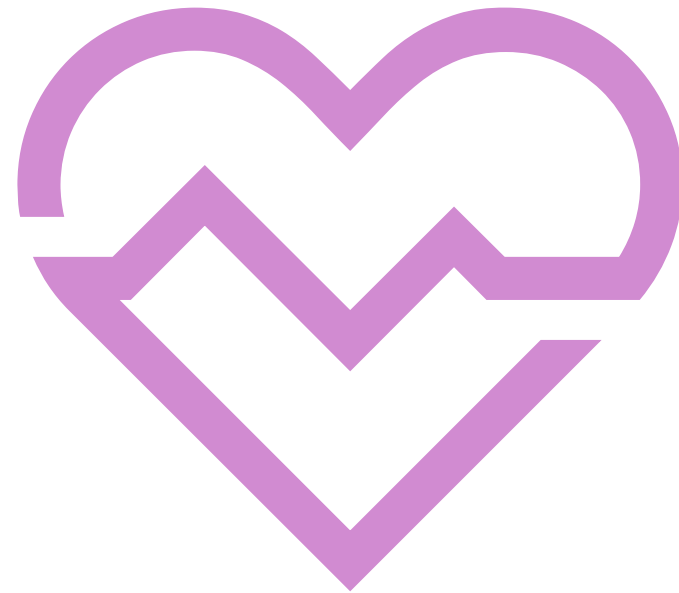
Fetal death



SIGNS AND SYMPTOMS PATIENT SHOULD WATCH DURING PREGNANCY FOR HYPERTENSION

Preeclampsia or Eclampsia symptoms:

- Swelling in face and hands
- Severe headache
- Vision Change
- Severe upper abdominal Pain
- Nausea or Vomiting
- Shortness of breath



MEDICATION: BEFORE PREGNANCY

Antihypertensive Medications for patient who plan to become pregnant or at risk of an unplanned pregnancy

The most acceptable safety profile medication with human safety data are:

Labetalol

Nifedipine Long Acting

Compatible with Pregnancy but with low effectiveness:

Methyldopa





MEDICATIONS: DURING PREGNANCY

1st line options:

Beta Blockers – **Labetalol**, Pindolol, Acebutolol, Metoprolol and Propranolol

Some beta-blockers have impaired fetal growth studies, but Labetalol consider safer due to it's Alpha and Beta Blocking activity. The other beta blocker also consider safe as they are listed above.

Long Acting Nifedipine also consider safe as it does have some studies with good outcomes.

Methyldopa has been considered safe for pregnancy, but it is less effective compared to Labetalol and Nifedipine LA

2nd and 3rd Line options:

Thiazides are not teratogenic but has little theoretical concern for intravascular volume depletion which can be restriction for fetal growth. If the patient was on thiazides before pregnancy, then it should be continued.

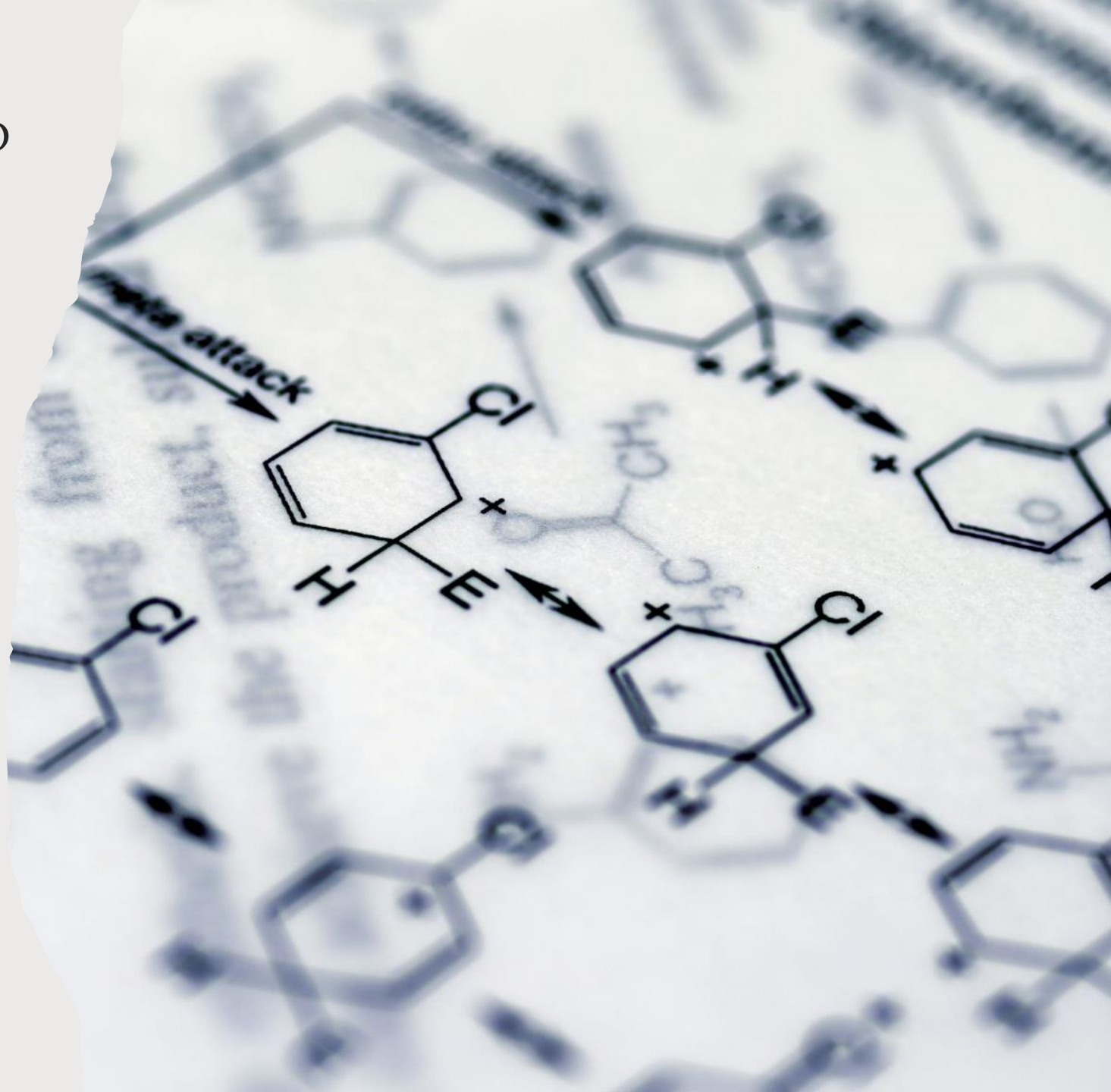
Clonidine and Hydralazine can also be considered as 2nd line option for treatment of HTN in pregnancy.

CONTRAINDICATED DRUGS DURING PREGNANCY

RAAS agents are contraindicated due to fetal kidney development issue.

ACEIs and ARBs are contraindicated from 2nd trimester due to various birth defects and underdeveloped organs.

Direct Renin Inhibitor **Aliskiren**, and Aldosterone Antagonists **Spironolactone, Eplerenone** are not recommended.



MEDICATION: DURING LACTATION

Beta-Blocker – **Labetalol**, **Metoprolol** and **Propranolol** reported to be safe and do not transfer to breast milk.

Do not use Beta- Blocker – **Atenolol**, **Acebutolol**, **Bisoprolol** and **Carvedilol** are not recommended.

Methyldopa consider to be safe during lactation period.

Calcium Channel Blocker – Amlodipine, Diltiazem, Nifedipine and Verapamil does transfer to breast milk in low amount, but still consider to be safe for babies 2 months or older.

Thiazide diuretics – Hydrochlorothiazide 50mg is compatible with breastfeeding, but it can decrease milk volume.

ACEIs – Captopril, Enalapril, Quinapril consider to be safe due to low amount transfer to breast milk, but with limited human data.

ARBs – Candesartan have low level accumulation in breast milk, therefore consider safe but no data available.

Spironolactone is also considered safe and compatible with breastfeeding.

CONCLUSION:

As we have seen what impact Hypertension can cause for pregnant mothers and fetus, therefore it is very important for us as a pharmacy care team to explore all the options for our young female patients to get start on or continue their High blood pressure therapy. We should continue consulting them regards to any adverse effect and let them know we are here for them to feel free to talk about planning a healthy pregnancy and what kind of care she will be needing during pregnancy and after delivering a health baby.

Thank You

Link:

